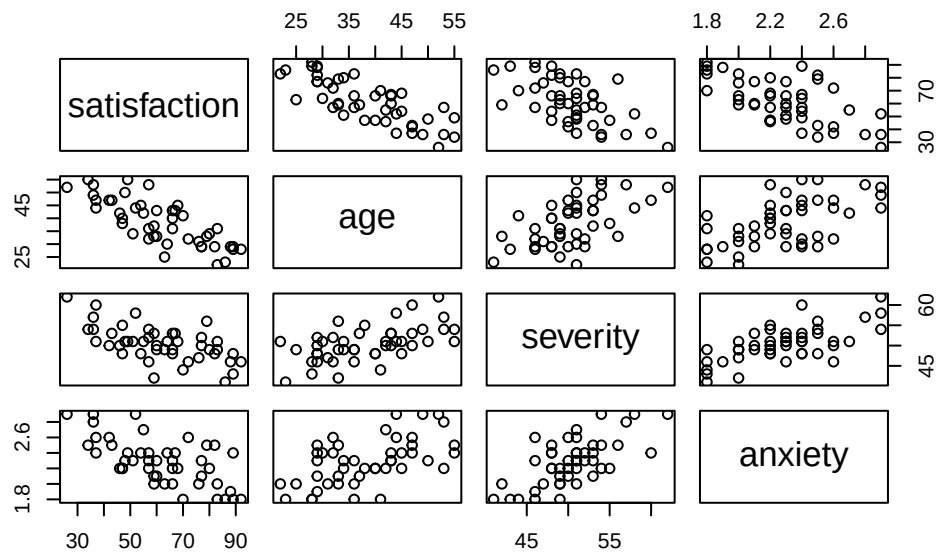


Lab 4

Brady Lamson

Part A

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pairs(data)
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3

Satisfaction is most highly correlated with age with a correlation of $-.787$

Highest negative correlation is satisfaction and age, highest positive correlation is severity and anxiety with a correlation of $.670$.

4

A

$$\hat{Y} = 158.49 - 1.14X_1 - 0.44X_2 - 13.47X_3$$

The value of b_1 indicates that satisfaction drops as age increases.

The value of b_2 indicates that satisfaction slightly drops as severity increases. Though this result is not statistically significant.

The value of b_3 indicates that satisfaction drastically drops as severity increases. Though this result is not statistically significant.

B

$$R^2 = 0.68$$

68% of the variation in satisfaction is explained by the model with age, severity and anxiety.

C

$$F = 30.05p = 1.542 \cdot 10^{-10}$$

As the p-value is far lower than any reasonable alpha, we have evidence to reject the null hypothesis that all of the slopes are equal to 0. There is a significant relationship between at least one of the predictors.

D

$$\sigma^2 = 10.06^2 = 101.2036\sigma = 10.06$$

E

$$s\{b_1\} = .2148s\{b_2\} = .4920s\{b_3\} = 7.10$$

F

B1 test:

$$t = -5.315p = 3.81 \cdot 10^{-6}yes$$

B2 test:

$$t = -.898p = .3741no$$

B3 test:

$$t = -1.897p = .0647no$$

5

I would say the assumption of normality of the error term appears to be reasonable.

6.

Yup, constant standard deviation seems to be okay here.

Part B

LengthCm and LengthIn are totally dependent on each other, they're the exact same measurement.

Part C

There isnt enough rows of data to compute these values. Too many predictors, not enough data.